Autoencoder vs PCA vs SVD Report

MinMaxScaler – Training Loss Curves

All architectures converge smoothly, indicating stable training. The [16] architecture achieves the lowest loss (~0.07), followed by [8]. Deeper architectures ([16,4], [8,2]) plateau at higher loss, showing that increased depth does not improve performance. Capacity (width) is more important than depth.

MinMaxScaler – Reconstruction MSE

Autoencoder performs best at latent size 16. PCA achieves very low error at high dimensions but degrades at lower ones. Autoencoders outperform PCA/SVD at low dimensions due to nonlinear representation learning.

MinMaxScaler – Training Time

The [16] model is both fastest and best performing. Deeper models are slower and generalize worse.

StandardScaler – Training Loss Curves

Similar convergence behavior but higher loss values overall. The [16] model remains best, while deeper models perform worse.

StandardScaler – Reconstruction MSE

Autoencoder again performs best at latent size 16. PCA remains strong at high dimensions, while SVD performs worst overall.

StandardScaler – Training Time

The [16] model is fastest and best. Deeper architectures significantly degrade performance.

Overall Conclusions

Shallow, wide autoencoders outperform deeper ones. PCA is competitive at high dimensions, but autoencoders dominate at low dimensions due to nonlinear learning. Results are consistent across scaling methods. Sensor and cyclic features exhibit nonlinear structure. Recommended approach: use autoencoders for nonlinear compression and PCA for simple linear reconstruction.